

Mohammed Sufiyan

Boston, MA (open to relocate) | [LinkedIn](#) | 857-263-1400
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EDUCATION

Boston University

MS in Applied data Analytics (GPA: 3.9)

Presidency University

B Tech in computer Science and Engineering (Data Science)

Boston, MA

Sep 2025 - Dec 2026

Bengaluru, India

Sep 2020 - Jul 2024

WORK EXPERIENCE

Boston University

Graduate Teaching Assistant - MET CS 688 (Section A1)

Boston, MA

Sep 2026 - Dec 2026

- Appointed as Teaching Assistant for **MET CS 688**, supporting course instruction, student questions, grading/feedback, and faculty coordination for a **35-student Fall 2026 section**.

Boston University

Graduate Research Assistant - Urban Heat Island / CleanCampus Project | [Portfolio](#) | [GitHub](#) |

Boston, MA

Jan 2026 - Aug 2026

[Live Demo](#)

- Built the **CleanCampus IoT monitoring dashboard** with **Node.js REST APIs**, **PostgreSQL**, **PM2/nginx**, **DigitalOcean** server, and **Leaflet** to ingest live sensor readings and visualize **BU campus heat patterns**.
- Improved field monitoring by modeling **5 core PostgreSQL tables** for sensors, gateways, zones, readings, and alerts, then adding filters, trend charts, alerts, and **CSV export**.
- Presented **BU Urban Heat Island/CleanCampus** research at **BU** and **ACSP 2026** in **Pittsburgh**, turning sensor data into clear planning visuals.

Boston University Housing

Summer/Fall Operations Assistant - StarRez / Housing Systems

Boston, MA

May 2026 - Present

- Standardized **BU Housing** operations by building a **DBMS-backed StarRez pipeline for room/furniture records**, summer assignments, **check-in/out updates**, and staff task tracking.
- Supported **full-time summer** and fall housing operations across all **BU dorms** by maintaining accurate **StarRez** records, **room readiness** updates, and resident/staff request workflows.

Power Tech

Data Engineer

Bengaluru, India

Jun 2024 - May 2025

- Reduced processing time by 40%** by automating 15+ Python data cleaning/transformation workflows across 3 internal datasets.
- Improved reporting performance by 30%** by optimizing 20+ SQL queries and building **5 Power BI dashboards** for daily operations reporting.
- Processed **10GB+** daily by building **AWS S3/Lambda ETL** pipelines and containerizing **4 apps** with **Docker**, cutting deployment time from **2 hours to under 30 minutes**.

ACADEMIC PROJECTS

FinGuard Real-Time Fraud Intelligence Platform | [GitHub](#)

Oct 2026

Engineered a **real-time fraud pipeline** with **Redpanda/Kafka**, Python producer/consumer services, **PostgreSQL**, **Redis**, **FastAPI**, **Streamlit**, and **Docker** to score card transactions and expose high-risk alerts through API/dashboard layers.

DataPulse AI Self-Healing Data Pipeline | [GitHub](#)

Oct 2026

Designed a **self-healing data quality workflow** with **Pandas**, **PostgreSQL**, **FastAPI**, **Streamlit**, and **Docker** to validate schema/type/domain rules, quarantine failed rows, score pipeline health, and generate repair suggestions.

PaperGraph Research Intelligence Engine | [GitHub](#)

Oct 2026

Created a **research intelligence system** that indexes AI/medical imaging papers in **PostgreSQL**, extracts **methods**, **datasets**, **metrics**, builds graph edges, and compares results through search and leaderboard APIs.

Agentic LLM Web QA System | [GitHub](#)

April 2026

Implemented a local **LangGraph/Ollama** QA workflow with **researcher**, **evaluator**, and **writer agents** to search the web, **rank JSON evidence**, and return sourced answers instead of raw model output.

Formula 1 Pit Stop Analytics DBMS | [GitHub](#)

Dec 2025

Modeled a **Formula 1 pit-stop analytics** warehouse with **SQL Server**, **ETL staging**, normalized **OLTP** tables, **star-schema reporting**, and **ML features** for slow pit-stop prediction.

Bizzlink Full-Stack Networking App | [GitHub](#)

Jan 2024

Developed a **full-stack** entrepreneur networking app with **Flutter**, **Node.js/Express**, **MySQL**, **Firebase** image storage, **authentication**, profile search, and connection workflows.

RESEARCH PAPER

Pix2Pix-GAN-Based Lung Segmentation for Chest X-Rays | [IEEE](#)

Sep 2024 - Apr 2025

Built a **Pix2Pix-GAN** lung segmentation workflow for chest X-rays to improve medical image preprocessing; authored **IEEE-published paper**.

LLMs in Reinforcement Learning: Comparative Analysis

Nov 2025 - Present

Working with a Machine Learning professor to compare **LLM-guided policy shaping** and reward design in **PPO-based** reinforcement learning; manuscript in progress.

SKILLS

Languages: Python, SQL, R, JavaScript, Node.js, Java

Data & ML: ETL, Data Modeling, DBMS, Statistical Analysis, Regression, ANOVA, Scikit-learn, PPO, LLMs

Databases: PostgreSQL, SQL Server, MySQL, MongoDB, Redis, Firebase

Cloud & Tools: AWS S3/Lambda, DigitalOcean, Docker, Kubernetes, PM2, nginx, REST APIs, StarRez, Git

Analytics: Power BI, Tableau, Excel, Leaflet, Dashboards, CSV Export, Data Visualization